				[5]			
16			$u = 1 + \sqrt{x}$	**B1**	**3.1a**	for use in substitution	No marks for attempts based solely on integration by parts
			$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}}$	**M1**	**1.1**	allow **M1** for $x^{-\frac{1}{2}}$	
			$\sqrt{x} = u - 1$	**M1**	**1.1**	allow sign error	
			$\int \frac{(u-1)^2 \times 2(u-1)}{u} \mathrm{d}u$	**A1**	**3.1a**	du may be seen later	
			$\frac{2(u^3 - 3u^2 + 3u - 1)}{u}$ soi	**M1**	**2.1**	allow sign errors and/or omission of 2	
			$2\left[\frac{u^3}{3} - \frac{3u^2}{2} + 3u - \ln u\right]$	**M1** **A1**	**3.1a** **1.1**	divides their cubic through by u and integrates dependent on award of first two **M** marks; allow sign errors and coefficient errors	must have $\ln u$
			$\frac{2(1+\sqrt{x})^3}{3} - 3(1+\sqrt{x})^2 + 6(1+\sqrt{x}) - 2\ln(1+\sqrt{x})$ $+c$ **oe isw**	**A1**	**3.2a**	if answer fully correct but either + c or du not seen then withhold final **A1**	
				[8]			

Question			Answer	Marks	AOs	Guidance	
			Alternative $x = u^2$	**B1**		*or* $u = \sqrt{x}$	may see $u = \frac{1}{1+\sqrt{x}}$ or $e^u = (1+)\sqrt{x}$
			$\frac{dx}{du} = 2u$	**M1**		allow **M1** for $\frac{du}{dx} = x^{-\frac{1}{2}}$	
			$\sqrt{x} = u$	**M1**		$x = u^2$	
			$\int \frac{u^2 \times 2u}{1+u} du$	**A1**			
			$(2u^2 - 2u + 2) - \frac{2}{1+u}$ from long division **oe**	**M1**		allow sign errors and/or omission of 2	
			$\frac{2u^3}{3} - u^2 + 2 - 2\ln(1+u)$	**M1** **A1**		integration attempted; allow sign errors and coefficient errors	
			$\frac{2x\sqrt{x}}{3} - x + 2\sqrt{x} - 2\ln\left(1+\sqrt{x}\right) + c$	**A1**		if answer fully correct but either + *c* or d*u* not seen then withhold final **A1**	
				[8]			